

Chaos Theory: 0/0 at the Onset of Chaos

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The Removable Singularity Framework

Abstract

We show that the onset of chaos in the logistic map is a 0/0 removable singularity. The Feigenbaum constants ($\delta=4.669\dots$, $\alpha=2.502\dots$) are UNIVERSAL -- the same for ALL maps with a quadratic maximum. This is the same universality as the Ising model ($\beta=1/8$) but for dynamical systems. Period doubling creates a fractal bifurcation diagram connecting to the Mandelbrot set. The Lyapunov exponent equals zero at the bifurcation point (0/0). Chaos connects to turbulence, SOC, finance, and consciousness.

1. The Logistic Map

$$x_{\{n+1\}} = r * x_n * (1 - x_n)$$

The logistic map is the simplest model of chaos. For $r < 1$: extinction. For $1 < r < 3$: stable fixed point. For $r > 3$: period doubling (2, 4, 8, 16, ...). At $r_{\text{inf}} = 3.5699\dots$: accumulation point (0/0). Above r_{inf} : chaos.

2. Feigenbaum Constants (Universal)

$$\delta = 4.669201609\dots \quad (\text{universal})$$

$$\alpha = 2.502907875\dots \quad (\text{universal})$$

The Feigenbaum constants are UNIVERSAL -- the same for ALL maps with a quadratic maximum: logistic map, sine map, any unimodal map. This is the same universality as the Ising model ($\beta=1/8$) but for dynamical systems. The constants were discovered by Mitchell Feigenbaum in 1975 and proved by Lanford in 1982.

3. Period-Doubling Route to Chaos

The period-doubling cascade: period 1 $\rightarrow$ 2 $\rightarrow$ 4 $\rightarrow$ 8 $\rightarrow$... The bifurcation points converge geometrically with ratio δ . At each bifurcation, the system becomes more complex. At r_{inf} , the period is infinite (0/0). Beyond r_{inf} : chaos.

4. Lyapunov Exponent

$$\lambda = \lim (1/n) * \sum (\ln |f'(x_i)|)$$

The Lyapunov exponent measures sensitivity to initial conditions. $\lambda < 0$: stable (periodic). $\lambda = 0$: bifurcation point (0/0). $\lambda > 0$: chaos. At the onset of chaos, $\lambda = 0$ exactly.

5. Universality: Same as Ising!

The Feigenbaum constants are universal, just like the Ising critical exponents. The logistic map and sine map have the SAME δ and α . This is because both have a quadratic maximum. Any map with a quadratic maximum has the same Feigenbaum constants.

- Ising: $\beta = 1/8$ (same for all Z2 systems)
- Feigenbaum: $\delta = 4.669\dots$ (same for all quadratic maps)
- Both: UNIVERSALITY from symmetry/universality class

6. Bifurcation Diagram: Fractal

The bifurcation diagram IS a fractal. Zoom into the diagram and find mini-copies of the whole structure. This connects to the Mandelbrot set (Chapter 42): both exhibit self-similarity and fractal structure at the onset of complexity.

7. Connections

Chaos	Prior Chapter	Connection
Fractal bifurcation	Ch.42 Fractal	Self-similar
Route to chaos	Ch.37 Turbulence	Period doubling
Self-organization	Ch.41 SOC	Dynamics
Market chaos	Ch.38 Finance	Price dynamics
Brain chaos	Ch.34 Consciousness	Neural chaos
Universality	Ch.36 Ising	Same concept

8. Conclusion

Chaos theory reveals that the onset of chaos is a 0/0 removable singularity. The Feigenbaum constants are universal, just like Ising critical exponents. The bifurcation diagram is fractal, connecting to the Mandelbrot set. The Lyapunov exponent equals zero at the bifurcation point. Chaos connects to turbulence, SOC, finance, and consciousness.

References

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